(a) Action-conditioned dual-branch co-autoregressive prediction

append prediction to history; repeat

RGB observations



source frame

MAGVIT-2

$s_{1:t-1}$

Actions a
↓ encoder E_a

c + position e (broadcast)

RGB ST block l
Spatial attention ↔
Temporal: causal →

RGB ST block L
Spatial attention ↔
Temporal: causal →

$\hat{s}_t$

Decoder

Next RGB frame

$\hat{o}_t$

RGB output

$\mathcal{L}_{\text{RGB}} = \sum_t \text{CE}(s_t, \hat{s}_t)$

Target: tokenized source RGB

Depth feature projection + RGB

W^l

Fusion at every paired ST block

W^L

Temporal depth annotations

Video Depth Anything

MAGVIT-2

$z_{1:t-1}$

Depth ST block l
Spatial attention ↔
Temporal: causal →

Depth ST block L
Spatial attention ↔
Temporal: causal →

$\hat{z}_t$

$\mathcal{L}_{\text{Depth}} = \sum_t \text{CE}(z_t, \hat{z}_t)$

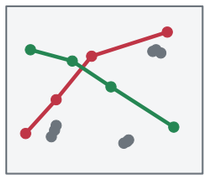
Target: tokenized annotated depth

Rollout: first RGB frame + actions → 15 future frames; losses and tracking are training-only.

predicted RGB tokens

(b) Adaptive keypoint dynamics — training only

Source video → SpatialTracker



Dense tracks (token coordinates)

$M_i = \sum_{t=1}^{T-1} \|p_i^{t+1} - p_i^t\|_2$

Select top-K motion magnitudes

Keep moving-point identities; discard small-motion tracks.

Trajectory sketches illustrate selection, not measured tracks.

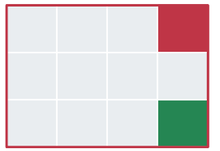
Gather predicted RGB tokens along selected tracks



$\hat{s}_1(p_i^1)$ $\hat{s}_t(p_i^t)$

$\mathcal{L}_{\text{Keypoint}} = \frac{\sum_{i=1}^{K}\sum_{t=2}^{T} \|\hat{s}_t(p_i^t) - \hat{s}_1(p_i^1)\|_2^2}{(T-1)K}$

Weight RGB token cross-entropy at tracked locations



$A_{t,x,y} = \gamma$ on tracks

$A_{t,x,y} = 1$ elsewhere

$\mathcal{L}_{\text{Attention}} = -\sum_t A_t \odot s_t \log p(\hat{s}_t)$

Loss weighting, not a Transformer attention matrix.

$$\mathcal{L} = \mathcal{L}_{\text{RGB}} + \lambda_1 \mathcal{L}_{\text{Depth}} + \lambda_2 \mathcal{L}_{\text{Keypoint}} + \lambda_3 \mathcal{L}_{\text{Attention}}$$

Jointly optimize the world model